BEING AN OPEN IMMERSION IS FPQC LOCAL ON THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The author's parenthetical '(this is obvious)' occurs in the proof and is quoted verbatim.

Editorial note. The counted spans carry navigation stubs for the references leaving them; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- f is universally open (resp. universally injective) when its base change along a flat surjective map of affines is. Same chapter, Lemmas lemma-descending-property-universally-open and lemma-descending-property-universally-injective, invoked at proof lines 6096-6099. Gives openness of the image and injectivity of f.
- A bijective homeomorphism of schemes is affine. Morphisms, Lemma morphisms-lemma-homeomorphism-affine, invoked at proof lines 6116-6117. Reduces the isomorphism case to a statement about rings.
- Base-change stability of open immersions, and the general principle for a property that is fpqc local on the base. Schemes, Lemma schemes-lemma-base-change-immersion and Same chapter, Lemma lemma-descending-properties-morphisms, invoked at proof lines 6085-6086 and 6128. Supplies the first half of the locality and concludes the proof.

Source reference. Original lemma 02KP: sources/stacks-descent/pinned-original-source.tex, source lines 5624–5624. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02KW: sources/stacks-descent/pinned-original-source.tex, source lines 5884–5884. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02KT: sources/stacks-descent/pinned-original-source.tex, source lines 5773–5773. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:descent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 04DE: sources/stacks-chow/tags.txt, source lines 5724–5724. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01JY: sources/stacks-chow/tags.txt, source lines 2031–2031. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *The property $\mathcal{P}(f) =$"f is an open immersion" is fpqc local on the base.*

Proof. The property of being an open immersion is stable under base change, see Schemes, Lemma 01JY. The property of being an open immersion is Zariski local on the base (this is obvious).

Let $S' \to S$ be a flat surjective morphism of affine schemes, and let $f : X \to S$ be a morphism. Assume that the base change $f' : X' \to S'$ is an open immersion. We claim that f is an open immersion. Then f' is universally open, and universally injective. Hence we conclude that f is universally open by Lemma 02KT, and universally injective by Lemma 02KW. In particular $f(X) \subset S$ is open. If for every affine open $U \subset f(X)$ we can prove that $f^{-1}(U) \to U$ is an isomorphism, then f is an open immersion and we're done. If $U' \subset S'$ denotes the inverse image of U, then $U' \to U$ is a faithfully flat morphism of affines and $(f')^{-1}(U') \to U'$ is an isomorphism (as $f'(X')$ contains U' by our choice of U). Thus we reduce to the case discussed in the next paragraph.

Let $S' \to S$ be a flat surjective morphism of affine schemes, let $f : X \to S$ be a morphism, and assume that the base change $f' : X' \to S'$ is an isomorphism. We have to show that f is an isomorphism also. It is clear that f is surjective, universally injective, and universally open (see arguments above for the last two). Hence f is bijective, i.e., f is a homeomorphism. Thus f is affine by Morphisms, Lemma 04DE. Since

$$\mathcal{O}(S') \to \mathcal{O}(X') = \mathcal{O}(S') \otimes_{\mathcal{O}(S)} \mathcal{O}(X)$$

is an isomorphism and since $\mathcal{O}(S) \to \mathcal{O}(S')$ is faithfully flat this implies that $\mathcal{O}(S) \to \mathcal{O}(X)$ is an isomorphism. Thus f is an isomorphism. This finishes the proof of the claim above. Therefore Lemma 02KP applies and we win. □